

Scoryst

Music engraving in Typst



github.com/bernsteining/scoryst · typst.app/universe/package/scoryst

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Scoryst — Music Engraving Plugin for Typst

Scoryst is a music notation engraving plugin for Typst. It wraps [Verovio](#) as a WebAssembly module, rendering music from multiple input formats directly into SVG embedded in your document.

Quick Start

Render inline ABC notation:

typst

```
#import "../pkg/scoryst.typ": score
#score("X:1\nM:4/4\nK:C\nCDEF|GABc|")
```



Or define a show rule to render ABC code blocks automatically:

typst

```
#import "../pkg/scoryst.typ": score
#show raw.where(lang: "abc"): it => score(it.text)

```abc
X:1
T:Ode to Joy
M:4/4
K:C
EEFG|GFED|CCDE|E2D2|
EEFG|GFED|CCDE|D2C2|
```
```

Ode to Joy



API Reference

score

typst

```
#score(
  data,          // string: music (ABC, MusicXML, MEI, Humdrum, EsAC, PAE, Volpiano, CMME)
  options: none, // dictionary: verovio options
  page: 1,       // int: page number to render
  ..args,       // forwarded to Typst's image() (width, height, fit, alt)
)
```

pages

typst

```
#let n = pages(data, options: none)
```

Returns the number of pages for the given music data and options. Useful to loop over pages of a multi-page score.

Verovio Options

Options are passed as a Typst dictionary. They map directly to [Verovio's toolkit options](https://book.verovio.org/toolkit-reference/toolkit-options.html).

| Common | | |
|------------------|------------|--|
| adjustPageHeight | true | Crop SVG height to content |
| adjustPageWidth | false | Crop SVG width to content |
| scale | 100 | Scale factor (percent) |
| font | "Leipzig" | Music font: Leipzig, Bravura, Gootville, Leland, Petaluma |
| inputFrom | "auto" | Format: auto, mei, musicxml, abc, humdrum, esac, pae, volpiano, cmme |
| pageWidth | 2100 | Page width (MEI units) |
| pageHeight | 2970 | Page height (MEI units) |
| pageMarginTop | 50 | Top margin |
| pageMarginBottom | 50 | Bottom margin |
| pageMarginLeft | 50 | Left margin |
| pageMarginRight | 50 | Right margin |
| landscape | false | Landscape orientation |
| breaks | "auto" | Line breaks: auto, line, encoded, none |
| condense | "auto" | Condense: auto, none, encoded |
| transpose | "" | Transpose (e.g. "M2" for major second up) |
| header | "auto" | Header: auto, none, encoded |
| footer | "auto" | Footer: auto, none, encoded |
| Layout | | |
| spacingStaff | 12 | Spacing between staves |
| spacingSystem | 12 | Spacing between systems |
| spacingLinear | 0.25 | Linear spacing factor |
| spacingNonLinear | 0.6 | Non-linear spacing factor |
| unit | 9 | Base unit size (half staff space) |
| stemWidth | 0.2 | Stem width |
| barLineWidth | 0.3 | Bar line width |
| staffLineWidth | 0.15 | Staff line width |
| lyricSize | 4.5 | Lyrics font size |
| hairpinSize | 3.0 | Hairpin height |
| SVG Output | | |
| svgViewBox | false | Use viewBox instead of width/height |
| svgRemoveXlink | false | Use href instead of xlink:href |
| svgBoundingBoxes | false | Add bounding box rects (debug) |
| removeIds | false | Strip element IDs from SVG |
| smuflTextFont | "embedded" | SMuFL text font: embedded, linked, none |

Full reference: <https://book.verovio.org/toolkit-reference/toolkit-options.html>

Music Fonts

Five [SMuFL](#)-compliant music fonts are available. Set the font with the font option:

typst

```
#score(data, options: (font: "Petaluma"))
```

Petaluma



Leland



Gootville



Bravura



Leipzig



Supported Input Formats

Verovio auto-detects the input format for ABC, MusicXML, MEI, Humdrum, and EsAC. For PAE, Volpiano, and CMME, pass `inputFrom` explicitly.

All the files used in the examples are available in the project's [Github](#).

Exporting from notation software

If you're working with any of these scoring software this table sums up each supported export format with their documentation linked.

| Software | MusicXML | MEI | ABC | Humdrum |
|-------------------------------|----------|-----|-----|---------|
| MuseScore | ✓ | ✓ | | |
| Finale | ✓ | | | |
| Sibelius | ✓ | | | |
| Dorico | ✓ | | | |
| LilyPond | ✓ | | | |
| Flat.io | ✓ | | | |
| Noteflight | ✓ | | | |
| ABC tools | ✓ | | ✓ | |
| Humdrum tools | ✓ | ✓ | ✓ | ✓ |
| MEI tools | ✓ | ✓ | | |

MusicXML is the universal interchange format — virtually all notation software can export it. For the best results, export as uncompressed MusicXML (`.musicxml` or `.xml`, not `.mxl`).

ABC Notation

[ABC specification](#) · [Download ABC files](#)

ABC is a compact text-based format popular for folk and classical melodies. Chord symbols are supported.

typst

```
#score(read("bach-prelude-cmaj.abc"))
```

Prelude in C Major, BWV 846

J.S. Bach



MusicXML

[MusicXML 4.0 specification](#) · [Download MusicXML files](#)

MusicXML is the standard interchange format for notation software. It supports grand staff, multiple voices, dynamics, and full score layout.

typst

```
#score(read("adagio.xml"))
```

Italy

<http://abc.musicaviva.com/tunes/vivaldi-antonio/vivaldi-bach-adagio-cm.abc>

Italy
<http://abc.musicaviva.com/tunes/vivaldi-antonio/vivaldi-bach-adagio-cm.abc>

42.00

4

8

11

14

17

MEI (Music Encoding Initiative)

[MEI Guidelines](#) · [Sample MEI encodings](#)

MEI is a rich XML-based format used in musicology, supporting lyrics, polyphonic textures, fermatas, and detailed editorial markup.

typst

```
#score(read("schubert.mei"))
```

Mässig.
205.

Am Brun - nen vor dem Tho - re, da steht ein Lin - den - baum; ich träumt' in sei - nem
Ich musst' auch heu - te wan - dern vor - bei in tie - fer Nacht, da hab' ich noch im
Die kal - ten Win - de blie - sen mir grad in's An - ge - sicht, der Hut flog mir vom

7

Schat - ten so man - chen sü - ssen Traum. Ich schnitt in sei - ne Rin - de so man - ches lie - be
Dun - kel die Au - gen zu - ge macht; Und sei - ne Zwei - ge rausch - ten, als rie - fen sie mir
Ko - pfe, ich wen - de - te mich nicht. Nun bin ich man - che Stun - de ent - fernt von je - nem

13

Wort; es zog in Freud' und Lei - de zu ihm mich im - mer fort, zu
zu: Komm her zu mir Ge - sel - le, hier find'st du dei - ne Ruh, hier
Ort, und im - mer hör' ich's rau - schen: du fän - dest Ru - he dort, du

Humdrum

[Humdrum User Guide](#) · [Download kern files](#)

Humdrum uses a tab-separated spine structure with ****kern** encoding for pitches and durations. Widely used in computational musicology.

typst

```
#score(read("sample-humdrum.krn"))
```

Aus meines Herzens Grunde

Johann Sebastian Bach
1685–1750

Soprano

Alto

Tenor

Bass

15

EsAC (Essen Associative Code)

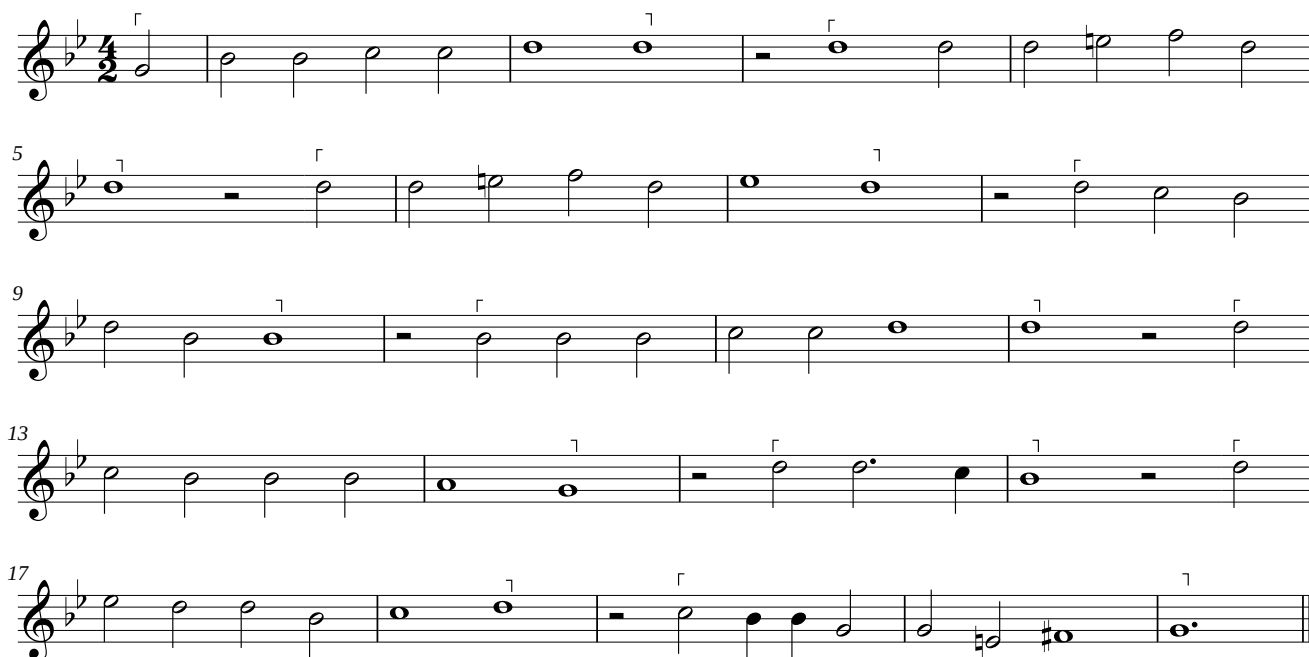
[EsAC data](#) · [Essen Folksong Collection](#)

EsAC is a text format used by the Essen Folksong Collection to encode melodies as scale degrees relative to a tonic. It stores metadata (title, region, key) and melody data in labeled fields. Here is *Das Hildebrandslied*, a German folk ballad. EsAC is auto-detected by Verovio.

typst

```
#score(read("hildebrandslied.esac"))
```

Das Hildebrandslied



Plaine & Easie Code (PAE)

[Plaine & Easie specification](#) · [RISM catalogue](#)

Plaine & Easie Code is a compact text notation used by [RISM](#) to catalogue musical incipits — the opening bars of a piece. It encodes clef, key signature, time signature, and note data as key-value pairs. Requires `inputFrom: "pae"`.

typst

```
#score(  
  "@clef:G-2\n@keysig:xF\n@timesig:3/8\n@data:=25//$xFCG @c 2-4.-'8E/{6AGFE}{8A'C}'B''4D{6C'B}/{''DC'BA}{''8EA}",  
  options: (inputFrom: "pae"),  
)
```



Volpiano

[Volpiano specification](#) · [CANTUS database](#)

Volpiano is a text encoding for medieval chant notation, used by the CANTUS database. Here is *Veni Creator Spiritus*, the famous Pentecost hymn. Requires `inputFrom: "volpiano"`.

typst

```
#score("1---g--h-ij---hgf--g--hg---k--lk--k7---hG--f---h--k--lk---l--m--l---k--lm---kj7--hg--kl---g--gh--k---jk---h---gf--h--hjh7---g--f--g7---3", options: (inputFrom: "volpiano"))
```



CMME

[CMME project](#) · [Download CMME files](#)

CMME is an XML format for mensural notation (medieval and Renaissance music). Requires `inputFrom: "cmme"`.

typst

```
#score(read("cmme.xml"), options: (inputFrom: "cmme"))
```

